

13. Pandas program to detect missing values (Display True/False)

```
import pandas as pd
```

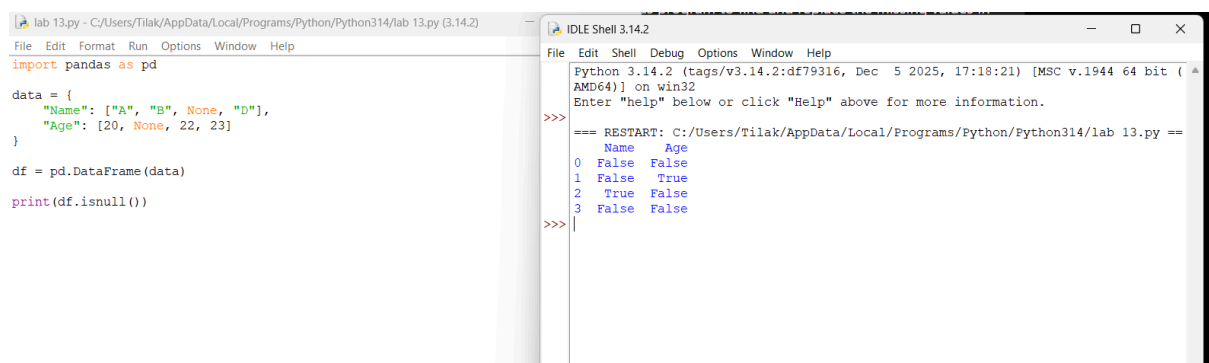
```
data = {  
    "Name": ["A", "B", None, "D"],  
    "Age": [20, None, 22, 23]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df.isnull())
```

Output:

	Name	Age
0	False	False
1	False	True
2	True	False
3	False	False



The screenshot shows a Python IDE with two windows. The left window, titled 'lab 13.py', contains the following code:

```
import pandas as pd  
  
data = {  
    "Name": ["A", "B", None, "D"],  
    "Age": [20, None, 22, 23]  
}  
  
df = pd.DataFrame(data)  
  
print(df.isnull())
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the code:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
>>>==== RESTART: C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 13.py ==  
      Name  Age  
0  False  False  
1  False   True  
2   True  False  
3  False  False  
>>>
```

14. Pandas program to find and replace the missing values

import pandas as pd

```
data = {  
    "Name": ["A", "B", None, "D"],  
    "Age": [20, None, 22, 23]  
}
```

```
df = pd.DataFrame(data)
```

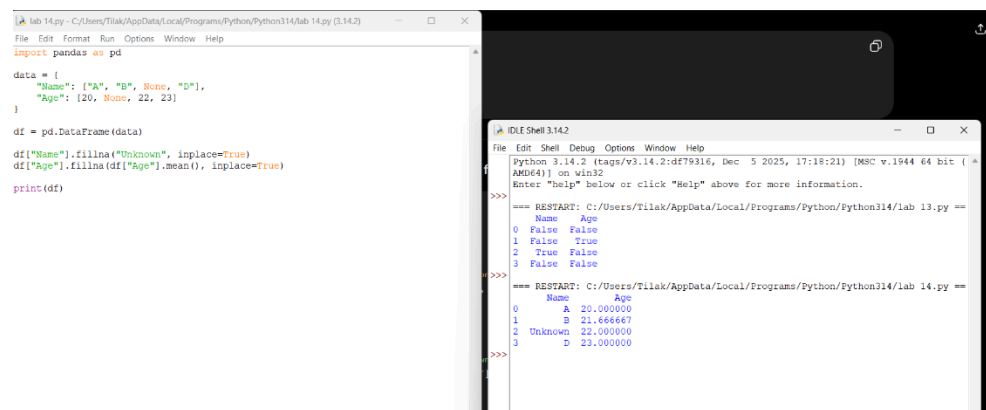
```
df["Name"].fillna("Unknown", inplace=True)
```

```
df["Age"].fillna(df["Age"].mean(), inplace=True)
```

```
print(df)
```

Output:

	Name	Age
0	A	20.000000
1	B	21.666667
2	Unknown	22.000000
3	D	23.000000



The screenshot shows a Python IDE with two windows. The left window displays the code for creating a DataFrame, filling missing values, and printing the result. The right window shows the output of the code, which is a DataFrame with 4 rows and 2 columns. The first row has Name 'A' and Age 20.000000. The second row has Name 'B' and Age 21.666667. The third row has Name 'Unknown' and Age 22.000000. The fourth row has Name 'D' and Age 23.000000.

```
lab 14.py - C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 14.py (3.14.2)  
File Edit Format Run Options Window Help  
import pandas as pd  
  
data = {  
    "Name": ["A", "B", None, "D"],  
    "Age": [20, None, 22, 23]  
}  
  
df = pd.DataFrame(data)  
df["Name"].fillna("Unknown", inplace=True)  
df["Age"].fillna(df["Age"].mean(), inplace=True)  
print(df)
```

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
  
>>> ===== RESTART: C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 13.py ==  
      Name  Age  
0  False  False  
1  False  True  
2   True  False  
3  False  False  
>>>  
  
===== RESTART: C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 14.py ==  
      Name  Age  
0      A  20.000000  
1      B  21.666667  
2  Unknown  22.000000  
3      D  23.000000  
>>>
```